Todo list

- Explicitly reference the ‘dependencies’ threads through the thesis rather than just saying ‘at many points’. 1
- DMR: I should frame this section as demonstrating the complexity of DMR signalling and why the transmission belt model is a useful solution as it doesn’t involve the same response gap or interference levels. 3
- Rather than 2D example: point out that it’s Hotelling-like, use the line example to show. 3
- Change the TBM 1st scenario table, it’s wrong. 7
- The paths not taken are perhaps just as important; this point is returned to in conclusion. 7
- Need to write up ‘dependencies’ clearly as a thread throughout. Tie to OOP section. . 7
- Have I intro’d bertrand comp somewhere else? X-ref. 9
- x-ref for ‘one day per day’ may change when I update framework chapter. 9
- Where have I mentioned the curve and magnitude parameters for smooth EoS 11
- This figure doesn’t quite fit! Move things, reduce the caption. 18
- Be sure to clarify what the markup value on the graph is: aiming for a marked up STOCK level. 18

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Chapter 1

‘Connected’ model results

1.1 Introduction

At many points in the thesis the discussion has been about the specific choices involved in creating a spatial economic agent model. It has been argued that many of these show strong interdependence: it is very hard to ‘decouple’ agents from each other and their environment. Choices made in designing supply impact directly on demand and vice versa. The way agents are able to communicate links them umbilically.

Explicitly reference the ‘dependencies’ threads through the thesis rather than just saying ‘at many points’.

This chapter brings these strands together in a model of the ‘transmission belt’ between production and welfare. It developed through the logical journey of building the model components. Some time will be spent discussing elements that, for reasons that will be explained, were rejected as possible routes to creating a ‘general’ model capable of fully linking supply and demand in a completely open geography where agents were able to make decisions separately.

Where this model succeeds best is in showing how otherwise murky dynamics can interact to produce spatial and production outcomes.

Section x demonstrates that it is possible for Firms to gain a ‘profit’ and that doing so has external effects on other Firms (as well as People). However, it does not find a way to create a competitive market structure through emergent Firm interaction. As will be explained, the specific route to achieving a ‘general’ model appears to rule out this kind of competitive dynamic. Instead, Firms can simply drive People down to next-to-zero utility.

So there are three key components out of four. One: the demand side works effectively, allowing People to make separate optimal buying choices that account for location and **love of variety**. Two: the supply side has a functioning production system, able to turn that demand into an equilibriating, stable economy that can shift between production regimes as key parameters change. Three: firms are able to gain a profit from the market. What it lacks is a forth key element: competition between Firms. There is a form of competition that can be seen when a

single Firm squeezes profit from the market: prices are interdependent and each price change creates an externality for other Firms. As one Firm lowers its price, thus increasing demand for its product, others’ price will rise as that demand shifts.

1.1.1 Organisation of the chapter

1.2 Moving towards more connected models

Producing partial models is a relatively straightforward task compared to getting Actors to fully coordinate economic activity. Of the various approaches attempted within this research, many were not able to recreate stable outcomes. Here are three quick examples. One: if the goal is to connect People’s labour to production, and thus to stock levels, but also to let Firms set good and wage prices, it is very difficult to get sensible stock levels. Section 1.3 presents a model that solves this stock problem, but it does it by forcing Actors to target a net-zero stock level. Firms using price-setting behaviour have a hard time producing stable economic outcomes: stock tends either to crash to zero or to increase indefinitely, without prices finding an efficient stock level. Section 1.2.1 presents a model designed to examine a key aspect of this problem: how Firms use the information they get to set their prices and how they can get it wrong.

Second, the problem of price signalling is thorny. The simple random money flow model in section **??** demonstrated a key part of the problem: treating money literally as a rivalrous good hits up against the tendency for the flow among many agents to ‘random walk’ into an uneven distribution. The model result even with Actors making more sophisticated maximising choices is that some Firms can end up with the entire money supply - more than they need - while others find themselves in the catch-22 of not being able to pay the wages to produce stock to sell to get money to pay wages. Any economic agent model faces being stuck between this naıve approach to money and finding some other method capable of signalling between Actors. The approach of section **??** is to cut out on layer of complexity by making Actors exchange ‘input’ (like their time) for productive output directly, where the price for the latter is the former. This avoids the need for an intermediary to signal price.

Lastly, it is also difficult to find any compromise position between partial and general models. For instance, Firms can be given unlimited stock to attempt to avoid the stock problems just discussed. A scenario using the Engel function illustrates the broader problem with this: if there is no stock limit, nothing stops Firms from pricing as close to zero as they can. The result is - in the case of the Engel function - luxury producers can price so low that the model reverts to the Cobb Douglas outcome of the share parameter dictating the split of revenue between sectors. Firms act rationally then - but a world of infinite stock makes a mockery of the idea of luxury and necessity goods.

Given this, section 1.3 presents a model that does manage a ‘general’ outcome, in that production is connected efficiently to welfare through price signals, and explains the methods used to do so. The next section examines one particular agent barrier to using price signals to

achieve coordination.

1.2.1 'Data, meaning, response' with two Firms

DMR: I should frame this section as demonstrating the complexity of DMR signalling and why the transmission belt model is a useful solution as it doesn't involve the same response gap or interference levels.

Rather than 2D example: point out that it's Hotelling-like, use the line example to show.

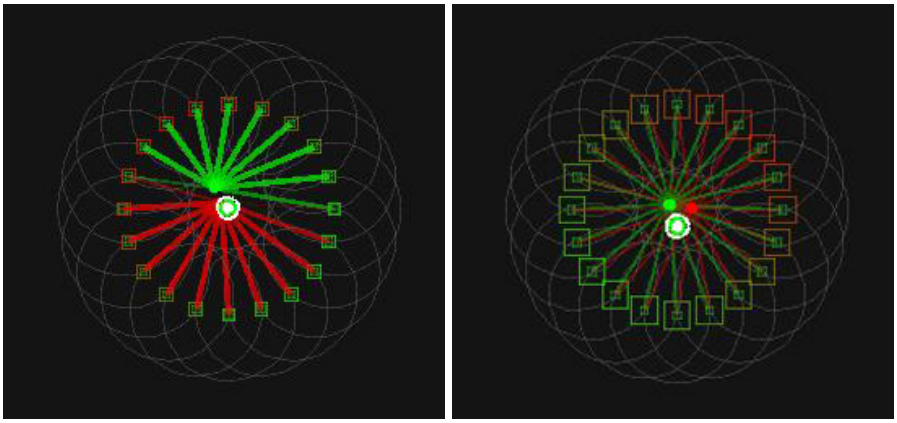
Component	**Description**
Run type	Parameter sweep, changing utility noise levels and changing random seed Single model run ends when Firms complete goal, see below.
Actor type	Two Firms, a small circle of People
Actor action	Firms: optimise location only People: spend set wage only
Timing	Randomised bloc for each bloc of Firms & People
Space-type	R^2
Actor location	Firms begin on edge, People fixed in circle
Space costs	Iceberg delivery costs for goods People pay this cost; Firms only receive good unit price
Parameters	All fixed, except utility noise level
Utility form	CES (love of variety)
Production form	None: demand always met

Table 1.1: Model components for 'Data, meaning, response' with two Firms'

A prerequisite for fully connecting production to welfare is Actors capable of successfully using cost signals to change their behaviour. Section 1.3 describes a scenario where these signals successfully lead to a higher-welfare outcome. The following example looks at a partial model to illustrate that the problem of cost signalling can be a challenge. Firms make only one decision: where to move? Getting closer to their market should, in theory, increase income - and indeed, if only a single Firm exists, it has no problem finding an income-maximising path. The rest of the world is static, so any change must be a consequence of its own actions. As soon as a second Firm is added, however, that is not the case. Confounding factors then mean they may fail to interpret the meaning of their cost signals correctly.

This model asks two Firms to play a 'game' on both a line and in R^2. Firms are placed an equal distance from a circle (or short line) of immobile People who are fixed at the centre within a small radius. The Firms can attempt to improve their revenue only by moving. Their revenue-maximising position is near the centre of the People, where delivery costs are minimised. A

game terminates either when both Firms have succeeded in getting inside the radius of the People or ten thousand days has passed. Figure 1.1 illustrates the endpoint of the game in R^2.



(a) $\rho = 0.995$ (b) $\rho = 0.5$

Figure 1.1: Two Firms at 'winning' position in centre of circle. Lines indicate buying strength from People: when goods are fully substitutable ($\rho \approx 1$) People buy from nearest Firm. When there is love of variety, they buy a mix from both.

Firms play an array of these games: utility noise levels are parameter-swept and for each value the random seed is altered a given number of times to then produce an average for all games with that value.

What is the rationale behind this? To put it into Data Meaning Response (DMR) terms (see section **??**) a Firm has only one datum to act on, a change in revenue. The meaning of an increase is taken to be, 'my previous action was successful'. Conversely, a drop in revenue is taken to mean that the action was counter-productive. This leads to the subsequent response: either repeat the action or reverse it. The broader meaning for the Firm in this situation is: 'my actions can determine my revenue levels'. Problems arise when this meaning is incorrect: a Firm may take an action, but if its revenue levels are not determined solely by that action, there is scope for misinterpretation, and thus a faulty response.

The aim is to provide a 'god's eye view' of the Firms' goal in as simple a cost-signal setting as possible. The optimum they are aiming for is clear. Firms having to get to a central point is, from the modeller's point of view, an entirely objective and clearly defined goal, while at the same time, opaque to the Firms, since their 'view' is restricted to revenue data. If they were not forced to use revenue alone to 'feel' their way, they could obviously be programmed to head straight for the centre of the circle. This makes it an easier example to consider than attempting to set prices. This combination - a single action, trying to optimise a single value - makes it useful for thinking about the problem. If Firms attempted to optimise other values (price or wage, for example) it would be considerably more opaque.

Although this looks like a Firm location model, it should not be thought of as one. Its purpose is to think about how Firms make decisions given (possibly noisy) information. Even putting aside that firms obviously do not 'grope' their way to more productive locations, they

would in reality be able to make much more informed location decisions. The problem here is more akin to having to make pricing decisions in a small monopolistic competition situation, where both other firms' and people's actions affect outcomes. The question is: in as simple an example as possible, what gets in the way of a Firm making an appropriate response to correctly interpreted data?

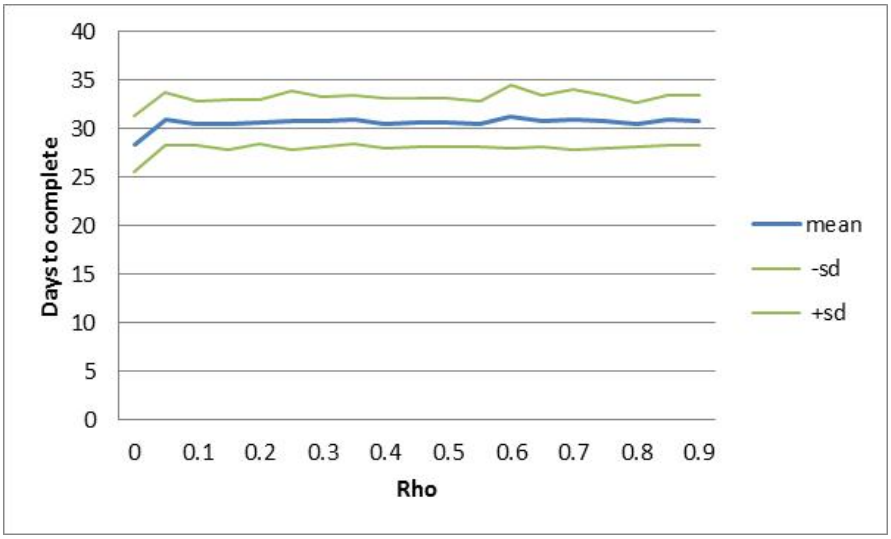
A full description of the setup for the 'game' test is as follows. People are set in a stationary circle at the centre of an R^2 space, or in a bounded line in R^1. Two Firms must find an optimal location that maximises their revenue. All prices and wages are fixed, and Firms' stock is unlimited. Delivery costs are 'iceberg': People pay a delivery cost, but it 'melts away' and Firms receive just the unit value. This means that, from the point of view of People, cost per unit drops as Firms draw closer. They can then afford to buy more. This means that a Firm's revenue-maximising location (in the absence of any others) is directly in the centre of the circle of people. A Firm's decision process is this:

- At time $t-1$: Try a new location near to my current one. If in R^1, this will be in one of the two available directions. If R^2, it will be a random vector defined by an angle and magnitude (as the magnitude is increased if the guess was successful; see section **??**).
- At t: did my revenue improve?
  - Yes: repeat the previous action, increasing the distance slightly
  - No: revert to last good location

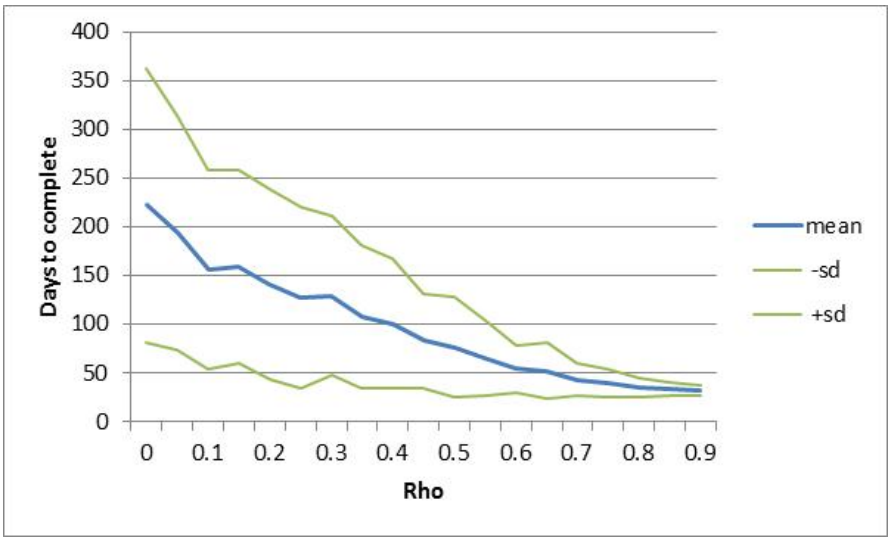
In this example there are five key factors that impact on revenue. The first is a Firm's own actions -just moving to a new spot. The second comes indirectly from the actions of other Firms: they may move to a location that leads People to shift demand to them, and this in turn may make other Firms' revenue-flow negative where it would otherwise have been positive. The other three factors all relate to the connection between People's demand levels and noise. If People are using pure constrained utility - thus providing a crystal-clear demand signal - Firms have no problems making correct decisions about changes in revenue. However, if demand is less clear than this from turn to turn, revenue signals become less reliable. More than this, as noise increases, the value of ρ - the level of love of variety in the model - begins to determine Firms' success rate.

In the model framework, noise naturally appears when using the greedy algorithm, and it is noisier the less iterations are used. As section **??** explained, the algorithm starts by guessing budget slices for all available goods. With fewer iterations, these guesses are further from the 'perfect' optimisation, and thus demand fluctuates. It is in running greedy algorithm models that the impact of noise on Firm optimisation becomes apparent. However, for the sake of computational expense and feasibility of large parameter sweeps, a noise term has been added to the constrained utility methods.[1]

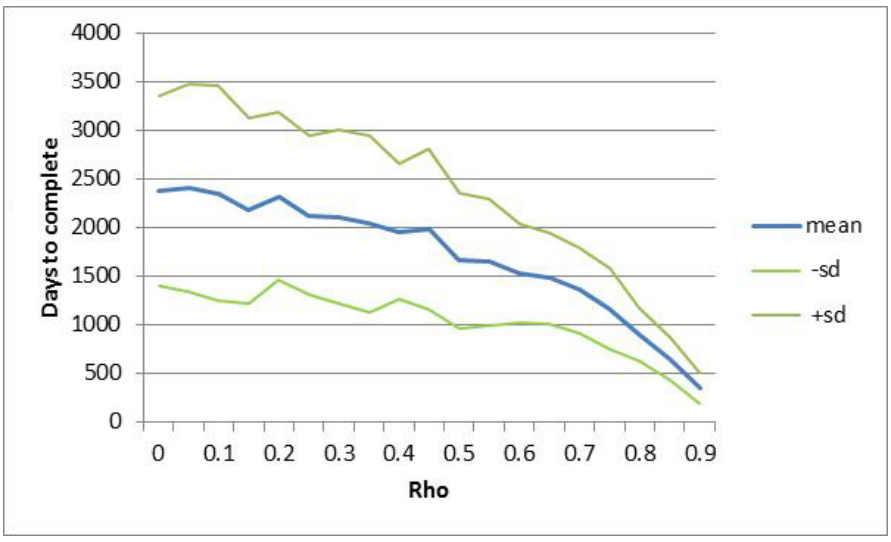
[1] While a noise term can be added to the constrained algorithm, it is quite hard to make it economically sensible.



(a) *noise* = 0



(b) *noise* = 0.1



(c) *noise* = 0.5

Figure 1.2: Firms compete to reach centre of market. Number of days to success on vertical axis, value of ρ on horizontal. Mean and S.D. for a hundred randomisations of each value point. Three different noise levels. Note for figure 1.2c, days to complete are an order of magnitude higher than for 1.2b.

The noise term just adds a random fluctuation to the optimal good quantities the constrained algorithm finds. For instance, if $noise = 0.5$, the optimal quantity is randomised in a range of plus or minus half its total value.

Firm and Person numbers are the final two factors, both impacting on demand noise level, if any is present. Both have a similar effect, for different reasons. Increasing the number of People means a larger source of demand for each Firm's goods, thus tending to average any noise out as these rise. Rather less intuitively, increasing Firm numbers reduces noise when using the greedy algorithm simply because the variance of size of budget slices drops. A budget randomly split between two Firms will have a much higher variance than one split between ten or a hundred. Constrained utility noise is equally lower, as it is applied proportionally to the quantity of optimised good amount.

Figure 1.2 graphs the average number of days it took both Firms to get inside the circle of People, for a range of values of ρ up to 1, in R^1. A hundred randomisations were averaged for each value of ρ. Three different noise levels were tested: it can be seen that when no noise is present, ρ makes no difference to the success of Firms' decisions. As the noise level increases, however, ρ begins to be important. High levels ($\rho \to 1$) present no problem, regardless of noise level. However, as love of variety increases ($\rho \to 0$) noise increasingly interferes in a Firms' ability to correctly identify revenue-improving positions.

The implications of this are discussed in more detail in the final chapter, in the context of the various approaches to coordination taken in the thesis.

1.3 The 'transmission belt' between production and welfare

Change the TBM 1st scenario table, it's wrong.

This section presents a model that connects supply and demand resulting in stable equilibria. Production with economies of scale are linked to consumers' welfare. It builds on concepts developed in the preceding analysis, finding a solution that allows efficiencies from economies of scale to feed through to improvements in utility. The features that distinguish the model are the result of the particular path the logical journey of model development has taken. It is as much the result of concepts that did not work as concepts that succeeded. It is *one* way to thread a needle through the many dependency problems the thesis has discussed. As the conclusion discusses in more detail, there are likely many other ways of making these dependencies function effectively together to produce other agent-based spatial equilibrium models.

The paths not taken are perhaps just as important; this point is returned to in conclusion.

Need to write up 'dependencies' clearly as a thread throughout. Tie to OOP section.

This requires staying in budget: the greedy algorithm does this by design. Adding noise to constrained utility, however, may mean People appearing to spend slightly too much or too little. Avoiding this is actually a large optimisation problem in itself, since there is no way to 'read back' from constrained good quantities.

Component	Description
Run type	Single run
Actor type	Single actor type only People number = 40 or 80
Actor action	Optimise utility, position, stock level
Space-type	None or R^1
Actor location	Mobile, random starting position
Space costs	Iceberg delivery costs for goods
Parameters	Each Person has 'given' exogenous time/input Each Person is also producer, has stock level & good price
Utility form	CES (love of variety)
Production form	smooth economies of scale

Table 1.2: Model components for 'love-of-variety production-welfare model'

The key features of the model are:

- People get goods directly as their wage. They buy goods directly with their time (a flow of 'one day per day') which then enters into Firms' production functions. Goods are sold using the 'goods plus delivery cost' equation from section **??**, $P_g = p_g + (c \cdot d)$. Goods and wage decisions are thus combined into a single utility decision.
- Firms charge base price p_g (the cost in a Person's time to buy one unit of good). Goods are exchanged directly for People's time, which then goes directly into the production function. Any good-movement costs 'melt away'.
- Production has economies of scale: higher input produces a higher per-unit output.
- Economies of scale work as described in section **??**: each Person 'leaves' their time with each Firm they contribute to until their next decision point. The production of stock is based on the amount of contributed time at each decision point.
- People use a Constant Elasticity of Substitution (CES) function to buy goods and distribute their time to Firms. This circumvents **Bertrand competition**-like problems with wage-setting, where a wage offer only minutely higher than others will automatically be selected. It means that each Person buys at least a little from each Firm - and thus contributes time to each - unless $\rho = 1$, in which case goods are fully substitutable and the cheapest will be bought.
- Demand for goods is always met: from the point of view of consumers, there are no stock limitations. The key advantage of this for the model is that a constrained utility

function can be used safely: a consumer buys a given amount based on the set of prices they observe. They face no risk of stock running out.

- Firms must then make sure that stock levels keep close to the amount being produced (or more than this amount if they are seeking to make a profit; see below). They do this by setting good costs to target a stock level - net zero stock if they are not attempting to skim a profit.

Have I intro'd bertrand comp somewhere else? X-ref.

x-ref for 'one day per day' may change when I update framework chapter.

In contrast to the two-timestep feedback decisions of Firms (used in section 1.2.1) stock levels are controlled through ongoing 'granular' price adjustments. This adjustment occurs every time a Person buys. It is a 'market mechanism' in that price signals are used by Firms to optimise a particular quantity though, as section 1.3.3 discusses, it is not a competitive one. The 'granular' process that occurs each time a Person buys a good is as follows:

- A Person chooses their best Bundle as described in section **??** using their unit of time as the budget. Since it is a CES function, all Firms will receive some share of this, minus the space cost that 'melts away'.
- When each Firm receives a signal that a Person is buying, it pays with the appropriate quantity of good based on its current price and stock is reduced accordingly.
- The quantity of input the Firm receives in return goes straight into production. The production amount is worked out as described in section **??**: economies of scale come from all input from recent interactions, and the actual productive output is calculated when the Person buys. It is proportional to the time that Person contributes. The method of achieving this is to use the smooth **economies of scale** function described in section **??**.
- There is now a net stock level from this one interaction, which the Firm uses to guide their price adjustment. If stock is more than zero, demand is lower than production, and so the price is lowered to try and sell more. If stock is less than zero, demand was higher than production, and so the price is increased to sell less.
- The change in price is proportional to the amount that stock levels deviate from the target amount: $\triangle p_g = f(stock)/x$, where x controls the magnitude of the price change. See section 1.3.2 for an explanation of the chosen function for stock targeting.

This section ends with some clarifications. Firstly, note that stock can go below zero for a time; indeed, this is vital for the price signal to target it from both sides. This zero target can be made physically 'plausible' by assuming Firms have stock buffers. Zero then becomes their

target production level rather than a physical lower limit. Indeed, the model can be set up to reach a given arbitrary level of production and then work from there, leaving a positive buffer. For the sake of ease of discussion, though, a target of zero is assumed (except for section 1.3.3 which uses the choice of stock target as a way for Firms to seek profit.)

Secondly, in contrast to the 'blind versus rational' problem described in section **??**, the transmission belt model *can* work with the fixed-cost, **core model** style of production function - but only if total input is able to exceed the fixed labour requirement. Otherwise, People buy goods that the Firm then cannot possibly make, since the production function becomes negative. Because of the fact that People 'leave' their time with Firms over an iteration, the fixed labour requirement can be met, but it leads to some nonsensical situations like the one just described. The smooth economies of scale function produces more reliable outcomes for the transmission belt model.

Lastly, a deeper explanation of the role of stock targeting in the model and its relation to efficiency (details of the actual mechanism are discussed below in section 1.3.2). The motivation was to find a way of linking labour input to good output and consumption. This requires all goods to be priced correctly. Many of the coordination problems involved in doing this have been discussed, especially the issue of getting time input to feed through to more efficient good production via economies of scale.

What does 'efficient' mean in this context? A given amount of time input into a Firm equates to a given level of good output via its production equation. The most efficient level of production from the point of view of People is where net stock from *all* Firms remains as close to zero as possible. Less than zero stock is both nonsensical in the long run and indicative that economies of scale are not being exploited as fully as they could be. Positive stock levels indicate that People are not getting all the available output that their labour is creating. The zero stock target is thus the point at which the benefits from production are maximised and feed through to utility levels.

This approach succeeds in finding efficient equilibria. In situations with many Firms and People, each Firm is able to find a unique price that keeps it own stock levels efficient while buying and selling to each Person. Their price signals find a balance point between production and People's utility demands as each Firms' price decisions interact with each other. (As explained below, the external effect of price changes are vital.)

The 'transmission belt' model can do some things the core model cannot but, in doing so, is also (as it is presently constructed) unable to do others. It is able to find an efficient production equilibrium given the number of Firms set in the model. This *does* succeed in linking efficient supply and demand through price signals. But it does not have a 'market structure' in the sense of firms competing for custom.

1.3.1 Defining the parameter space

Total production in a transmission belt model run is dependent on three things. One: the number of People, each having Distance EoS

1.3.2 The stock targeting mechanism

Component	Description
Run type	Single run
Actor type	One Firm, one Person
Actor action	Person: optimise utility Firm: target zero stock level
Space-type	None
Parameters	Each Person has ‘given’ exogenous time/input Firms use ‘smooth’ production function with economies of scale with curve $= 1$, magnitude $= 1$
Utility form	Linear since only one Firm

Table 1.3: Components for single Firm / single Person ‘transmission belt’ model

This section uses one Firm and one Person to demonstrate how the transmission belt model works. In the case of only one Firm, there is a very straightforward deduction of the correct price for ‘clearing the market’ and this can be compared with the model outcome. If there is only one Firm, all input goes into that single Firm: People have nowhere else to buy from and will spend everything they have there. So (assuming no space costs) the Firm’s output is directly proportion to the number of People, since each Person contributes a single unit of input. A Firm wants to clear its stock - to keep it as close to zero as possible. What price can it set to do this? Price is the output a Firm gives for a unit of input. People pay with a unit of input each so, in this simple case, price is always just the number of people divided by total output. Using the smooth economies of scale function and setting both the curve and magnitude parameters to one, output is just $t^2 + t$ (where t is the total input from all People). In cases where only one Person is contributing one unit of t to each Firm, the price equation reduces to $P = 1/t + 1$. In the case of one Firm and one Person, then, at equilibrium $P = 0.5$.

Where have I mentioned the curve and magnitude parameters for smooth EoS

In model runs with no distance cost, deducing the equilibrium price is straightforward for other mixes of Firm and Person number, as long as demand is either spread evenly or being input into a discrete subset of Firms, as happens when one Firm emerges as a selected site for large-scale efficient production (see section 1.3.4 below). If demand is spread evenly, output and price is the same as above. If, for example, two hundred people input into fifty Firms evenly,

that is four People's time per Firm. Equally, if one efficient producer emerges, all fifty People input their time into that Firm's smooth economies of scale equation.

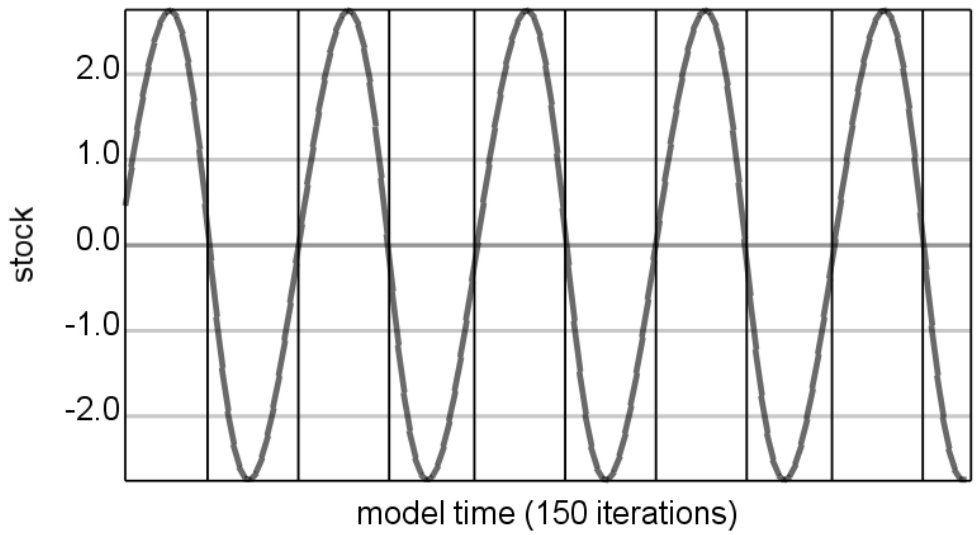
As explained, in the transmission belt model, each Firm adjusts their price in order to hit a set stock target. This adjustment is 'granular': it takes place at the exact moment any Person buys goods. There are problems with the crudest form of this mechanism, however: a mismatch between rates of change for price and stock levels. Using the simplest single-Actor example, this section also illustrates this problem and a solution that uses damping to more effectively equilibriate stock levels using prices.

Earlier iterations of the model used Firms that targeted zero stock only. This works under many circumstances, particularly with many agents as interaction effects act as a dampener on the control mechanism. However, using price as a control to target only zero stock means that, when stock hits zero, price will probably be in the wrong position to keep it there. Stock may well still be on a trajectory beyond zero - or, to frame this more usefully, if the first derivative of stock is not also zero (if it is still in the process of changing) price will have to swing back again. Figure 1.3 illustrates this using a single Firm and Person. As there is only a single producer and good, love of variety has no effect. There are also no delivery costs. In this example, a perfectly stable oscillation results as price adjustment and stock levels are always exactly out of sync.

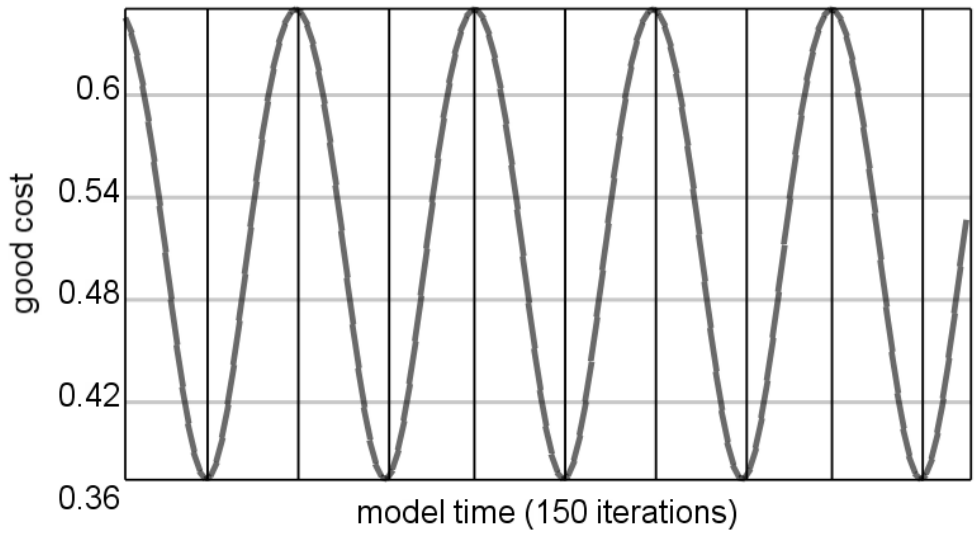
This is a version of a classic problem of harmonic oscillators that can be addressed by using damping (see e.g. Taylor 2005 pp.172). The goal is not only to target zero stock, but to aim for both stock and price rates of change to be zero at the same point. The solution is relatively straightforward: target the first derivative of stock as well as raw stock levels. If price is made to aim for the point where both are zero simultaneously, equilibrium should be reached. In fact, adding a second derivative zero target can improve the targeting even further.

Figure 1.4 shows three model runs, this time using these extra first and second order targets. The 'derivatives' are discrete differences in the model between timesteps. The first order derivative is the difference between stock at time $t-1$ and at t and the second order is the difference between this first order result between timesteps. The target mechanism itself does not change: if $s+s'+s'' \neq 0$, price is changed in proportion to how far from zero it is, as described above: $\triangle p_g = (s+s'+s'')/x$ where x controls the price change magnitude. If $s+s'+s'' > 0$, price is decreased and more stock sells. If $s+s'+s'' < 0$, price is increased and less is sold.

Sub-figure 1.4a shows good cost for a single Firm / single Person run. Sub-figure 1.4b shows good cost for fifty Firms with fifty People, with initial good costs randomised between 0.1 and 1.5. Sub-figure 1.4c adds a distance cost to those hundred agents. It gives them random, static locations in R^2 so that People buy from Firms at a range of distances, with each position being unique. In the two runs without distance cost, the equilibrium price of 0.5 is reached. It appears faster for sub-figure 1.4b (despite showing fewer iterations) because of the larger number of People: each time one of those People buys, it triggers a Firm's price response, so there are actually fifty such responses per Firm per timestep.



(a) Stock



(b) Good cost

Figure 1.3: Transmission belt model: a single Firm changing price to target zero stock. Vertical lines indicate where stock crosses the y axis. As stock hits zero, however, price is far from its equilibrium value and both stock and prices oscillate.

Sub-figure 1.4c introduces an element of distance: actors' locations are randomised on an R^2 plane of one unit distance per side and goods are given a delivery cost of 0.25. The result illustrates a basic point about the transmission model: a stable equilibrium can be found with agents facing heterogenous costs from different spatial locations. The result is a spread of equilibrium good prices, as each Firm receives a different quantity of inputted time, and thus the equilibrium production level to maintain the zero stock target is different for each. Figure 1.5 shows this result in more detail by correlating each Firm's good cost to the amount of time being contributed to it. This graph shows data from day 600, which is the very right hand point of sub-figure 1.4c, at a point of stability. For a Firm with a production function exhibiting economies of scale, higher input levels increase per-unit output levels. In seeking to maintain a net-zero level of stock, lower prices are found to clear this higher-efficiency level of production - with the effect also of increasing Peoples' overall utility levels.

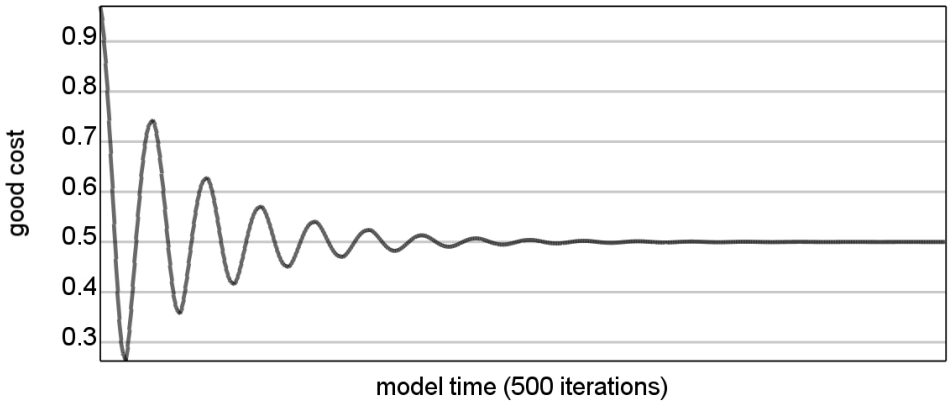
The classic equation for oscillation damping includes a term for angular frequency (ibid). Combined with a scaling parameter, in theory, the level of damping can be exactly targeted. The transmission model is 'under-damped' - it oscillates before reaching equilibrium. Addressing this, however, would require detecting the angular frequency of those oscillations. While it is relatively straightforward to detect inflection points for many model runs, other runs produce complex interaction effects that make this detection process unreliable. The solution used here is simply to manually set the scale of price reaction to stock changes for each model run using x, but it should be noted that this can impact on the exact point where Firms lose their grip on effective price-setting, as it determines the size of price swings - and consequently the scope for entering a positive feedback loop where stock targets are missed (see the next section).

This example draws attention to another important feature of the transmission belt model. With a single Firm (and assuming space costs stay constant) production levels cannot change. People will always spend their entire budget in the process of maximising their utility. As this budget is also the total time available for production, this also does not change. With a single Firm, this means its production level is fixed: varying price only affects the amount of input that is bought. With more than one Firm, changing price *can* effect production levels - but this is due to an externality. If one Firm increases its price, this makes other Firms' goods relatively cheaper: demand increases for them. Input thus shifts to those Firms.

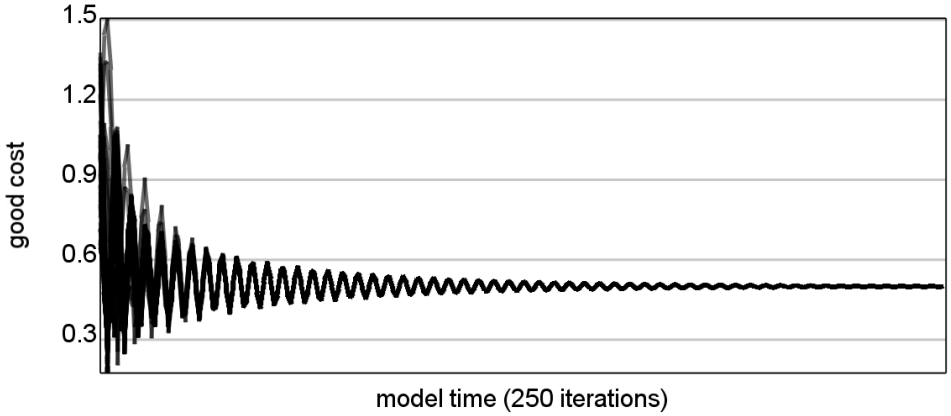
1.3.3 Markup

In the core model, a Firm's profit (or 'markup') does not exist as a variable in the economy. It has an implicit presence as the signal to other firms about the level of demand: it is argued that, if profits are above or below zero, firms enter and exit the market until profits are driven back to zero again. A change in firm number is the *only* way that production structure changes in the core model and even then, the actual number plays no direct role in deducing welfare outcomes.

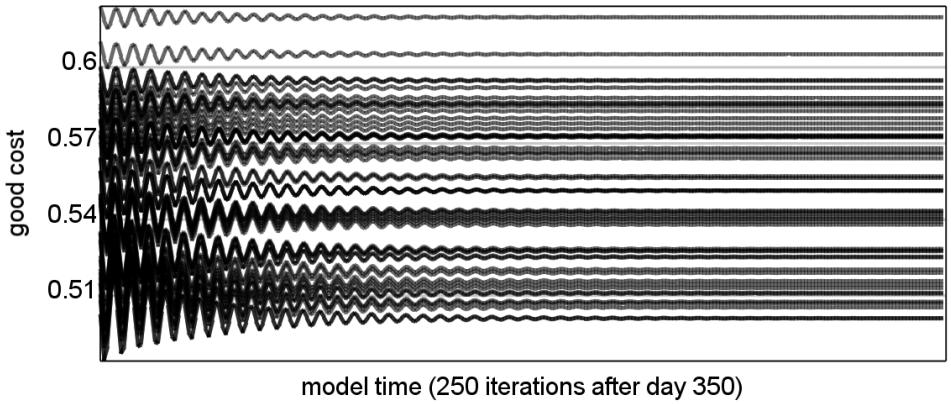
This chapter does not examine Firm entry and exit: the number of Firms is set exogenously. It is potentially a fruitful avenue of investigation as it offers another way to examine changes in



(a) Good cost for One Firm, one Person buying



(b) Good costs for fifty Firms, fifty People buying



(c) Good costs for fifty Firms, fifty People buying, delivery cost = 0.25, random locations in an R^2 plane of 1 unit distance to a side

Figure 1.4: Transmission belt model: targeting the first and second derivative of stock as well as zero stock itself, ‘damping’ the oscillations and achieving equilibrium.

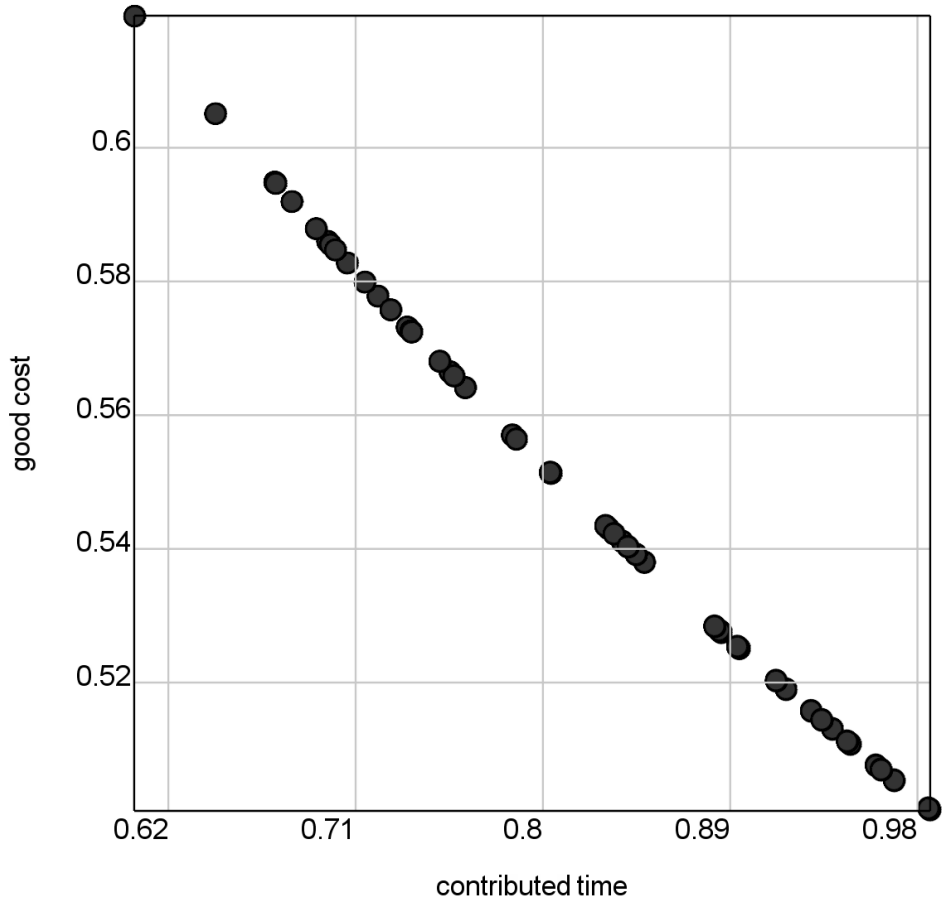


Figure 1.5: Transmission belt model: fifty Firms, fifty people buying, delivery cost = 0.25, random locations in R^2. Correlation of time contributed to each Firm versus their good cost at equilibrium, on day 600.

production structure. However, there are good reasons for leaving it out at this stage. Firstly, the models within this thesis have required agents to utilise only information they can collect from their environment. For a Firm to enter the market, having spied others making excess profits, it would need to be able to access that data directly. To know from a given price what profit a Firm is making requires also knowing their production output and markup. Starting to give agents this level of access to information takes the models into a different realm of knowledge and DMR. It would not necessarily require a ‘master agent’ watching global variables and adding or removing Firms appropriately; as section **??** discussed, an evolutionary approach might add Firms randomly and allow them to attempt survival, while also allowing failed Firms (see section 1.3.3) to ‘die’. But the mechanics of entry and exit are complex, particularly for the transmission belt model where any change can cause perturbations large enough to knock other Firms over a ‘tipping point’ (see below), causing runaway problems for the model dynamics.

In the transmission belt model, Firms *are* able to gain a profit - thus the first ingredient for a Firm-level economic micro-motive is present. But for this to provide Firms with complete economic motivation, they would need to be able to maximise that profit. In the core model, profit maximisation involves simply finding the first order zero-point of the profit equation. In the transmission belt model, some explicit mechanism is required. This section explains how Firms in the transmission belt model can gain a markup and how that causes external effects. But it also outlines that, as the model stands, creating an effective competition mechanism is challenging.

Firms can make a 'markup' on the price they charge, keeping an amount of output for themselves. But the dynamic created is a ratchet effect: each individual Firm is able to take extra slices of production for themselves, right up to the point where People are no longer able to afford to buy the remaining output that Firms do not keep as profit - at which point, there is no incentive to work and the system collapses. Note again that People have no exogenous options for spending their time and thus no effective sanction that could produce competitive behaviour in Firms. Even if goods' cost is exhorbitant, they will pay it, as no other options are available to their utility functions and a sliver of utility is better than none. They are still maximising.

Nevertheless, the mechanism that a Firm can use for gaining markup is worth setting out, as it differs from the traditional idea of setting prices directly. Instead, a Firm sets a target amount of output they want to skim from each transaction and then use price as a secondary targeting mechanism to reach it. Achieving this is slightly awkward, as attaining a markup means that extra output must be produced *constantly*. Consider: if a Firm simply aims for a higher stock target level and stops there, it is still net zero change over time. In order to get a mark up, it needs stock to be increasing over time, and to take that increase for itself. In order to achieve this, the Firm needs a markup level it aims for. This is then added to a moving markup target so that the Firm aims to skim a constant amount in a given time period. The stock target equation is altered to: $s + s' + s'' = mt$ where mt is the moving markup target[2].

Firms can easily succeed in achieving a stable markup over time. But if they attempt to set it too high, they fail, reaching a disastrous 'tipping point'. As figure **??** shows, equilibrium good prices for Firms seeking markup are higher than the zero-stock clearing price. This higher price emerges as it targets the higher level: it ends up charging extra and keeping a cut of the output. The external effect of this raised price is to drive demand to the second Firm. As that also increases input into that Firm's production, it must lower its price to sell the extra output it is making in order to hit its zero-stock target.

Ah: you need EoS for there to be an external effect on other Firms' prices, which is what Krugman was implying. Otherwise a change in input is linear with output change and the stock clearing price is the same.

What causes the tipping point? The Firm attempting markup reaches a price where it can

[2]There are slight alterations to this in the code, as it is easier to continue to target zero stock while shifting other values, but the principle remains the same.

no longer approach its stock target. The inability to move towards it only reinforces its price changes, and it enters a positive feedback loop: further price increases lock in its unaffordability and drive production to the other Firm. Figure 1.6 makes clear an important feature of this tipping point: it correlates to ρ. Why is this? ρ controls elasticity of substitution, thus the effect of any change on price. For higher values of ρ, as goods become more substitutable, People are less reliant on love of variety: a change in price becomes a stronger signal for the other Firm. In this model run, ρ is swept from $0.1 < \rho < 1$. One of two Firms slowly increases the markup it aims for. At a certain price, it reaches the tipping point. Each increment of ρ is triggered when this happens: the model is reset and run again, and one Firm again attempts to incrementally seek profit.

One of the two Firms (Firm zero) slowly increases its markup until it hits a 'tipping point' where it can no longer effectively target a stock level and its price enters a positive feedback loop. This is taken by the model as the signal to parameter-sweep ρ: it resets the Firms and runs again, for values of $0.1 < \rho < 1$. The stable markup value achieved before the feedback starts is graphed against ρ.

This figure doesn't quite fit! Move things, reduce the caption.

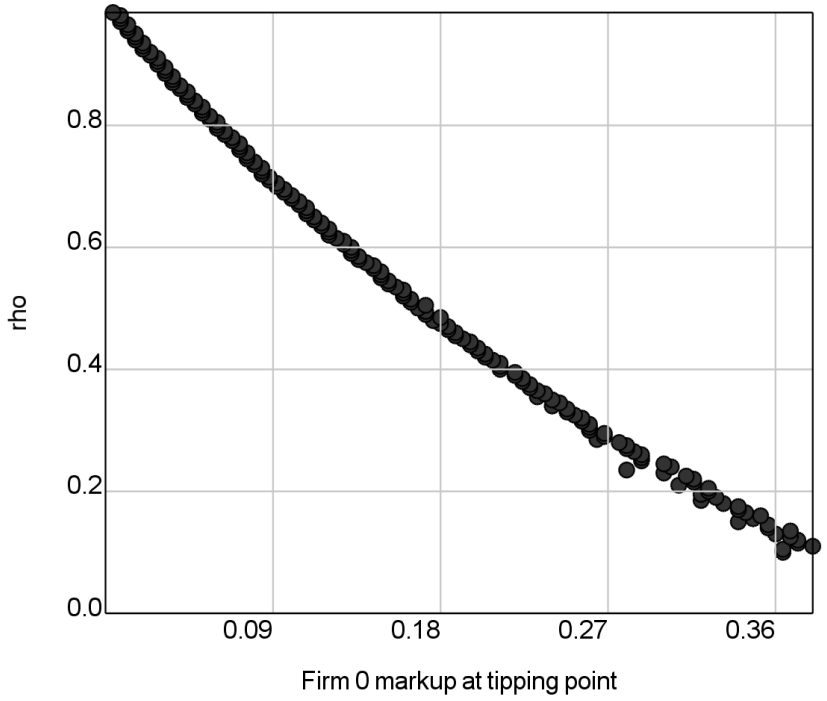
Be sure to clarify what the markup value on the graph is: aiming for a marked up STOCK level.

So there are only two possible outcomes for Firms attempting to seek profit, as the model is currently structured. If Firms collude, or succeed in following a collusion-like pattern through independent actions (implicit collusion), they can squeeze profit out of People right up to the point where People can no longer afford to buy anything at all. Or, if one or more Firms attempts to get markup, this is sustainable up to a point, but then tips over into a positive feedback loop between the external effect of lowered prices for others and the profit-seeking Firm's inability to use price to target their new stock level due to having lost too much demand. Missing their target causes them to continually increase their price, permanently compounding their error.

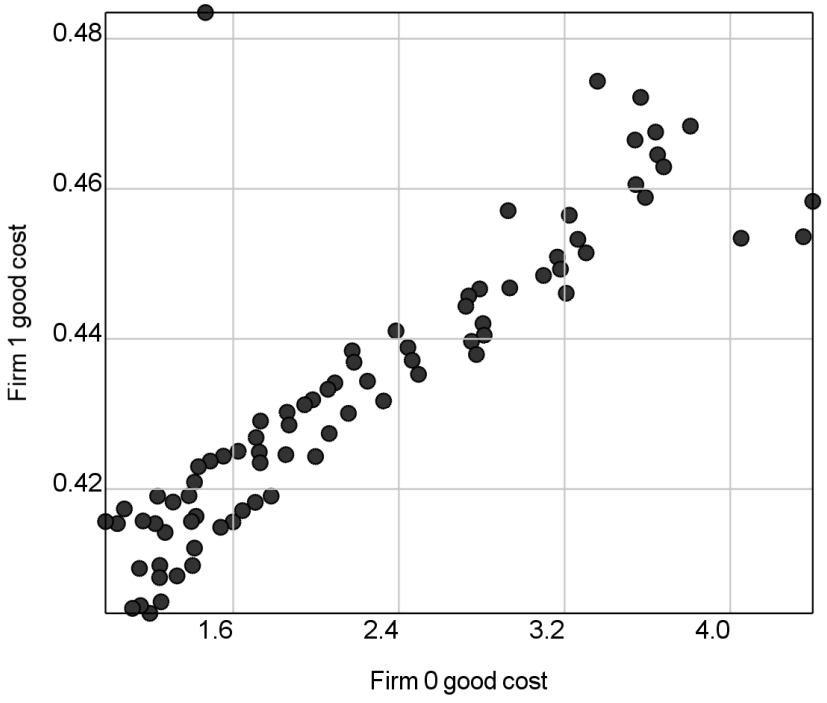
1.3.4 Interaction between economies of scale and love of variety

Section **??** explained that, in the **Dixit Stiglitz model** model, economies of scale at production level are subsumed into two things: the love of variety parameter (or its mirror, the elasticity of substitution parameter ε) and a market structure argument that reduces economies of scale to the entry and exit of firms. It is a very powerful set of deductions that allow everything else that follows a solid mathematical foundation. As Brakman *et al* (2009 p.106) note, " internal economies of scale are not absent... but show up in only in a rather special way." The core model does contain parameters for production level economies of scale in its 'scaffold', but the final model is not sensitive to their value in any way. Indeed, with the normalisations mentioned in section **??**, they are analysed away.

The transmission belt model uses explicit economies of scale at the production level while



(a) Firm zero seeks markup at varying values of ρ; no economies of scale



(b) Firm zero seeks markup; economies of scale cause external price effect. Two firms’ prices correlated.

Figure 1.6: Two Firm / two person transmission belt model. The first figure correlates Firm zero’s price as it raises its markup target. Data is shown at the tipping points, correlated to ρ; there are no economies of scale. The second includes economies of scale and shows the external effect of Firm zero’s price changes on Firm one’s price, driving it down.

consumers have an entirely separate love of variety utility function. This allows interaction between them to be investigated. This section demonstrates that, in the transmission model, the interaction of economies of scale into love of variety can determine the whole model’s production structure. Within the parameter space defined by economies of scale and love of variety, actors can find two distinct production regimes. In one, love of variety dominates and production is spread out among many producers symmetrically. In the other, the economies of scale effect dominates, and that symmetry is broken. While the transition is ‘catastrophic’, changes to relative production scale exist beyond that point: the state of the model is not simply binary.

The transition between these two regimes can happen in a model without space, as this section demonstrates by building on the same elements in the previous two sections.

1.3.5 Introducing space to the transmission model

May also require some ”static” demand a la agriculture sector - though may not. Depends if Firms can move. They can’t - they have no separate economic motivation to do so. Their stock target can’t help them. I can lump them together (a Firm follows a Person)... which we could show can lead to the core-periphery dynamic with just EoS. That’s an important point about market structure there: the same micromotive should drive price setting as well as location setting. Without it, the model is incomplete. I’m down with saying that. I also want to say: I don’t want to minimise the market structure argument made for the core model firms. It may be implicit but it is ”real” all the same, cf. Fowler. Given all that, what do I gain from making a two-region version that includes Person movement? If manufacturing cannot agglomerate, what are we looking at, even? Well: if we leave manufacturing evenly split, agglomeration can still occur through EoS alone. Again, there’s that caveat: what would the market structure implications be? One might speculate that if a Firm has lock-in, no market entry is possible.

Bibliography

Brakman, S., Garretsen, H., van Marrewijk, C. (2009). *The New Introduction to Geographical Economics*, 2nd Edition. Cambridge University Press.

Taylor, J. R. (2005). *Classical Mechanics*. University Science Books.